Mid-term (🙀) evaluations

Optional *anonymous* **survey** (not a quiz) available on Canvas all of **Week 5** (**next** week)

Three questions

- What do you like the most about how the course has been done so far?
- What do you like the least about how the course has been done so far?
- How can I help you learn better?

A note on quiz grades

- You need to provide the citation: Lecture # and Slide # or Reading and page #.
  - Citing your sources is standard practice in academic writing.
  - When a question asks for evidence, ***stick to evidence discussed in class.***

- Pulling citations from a lecture slide that we didn't actually read for this course doesn't count.
  - Next time I will add this instruction to every question.

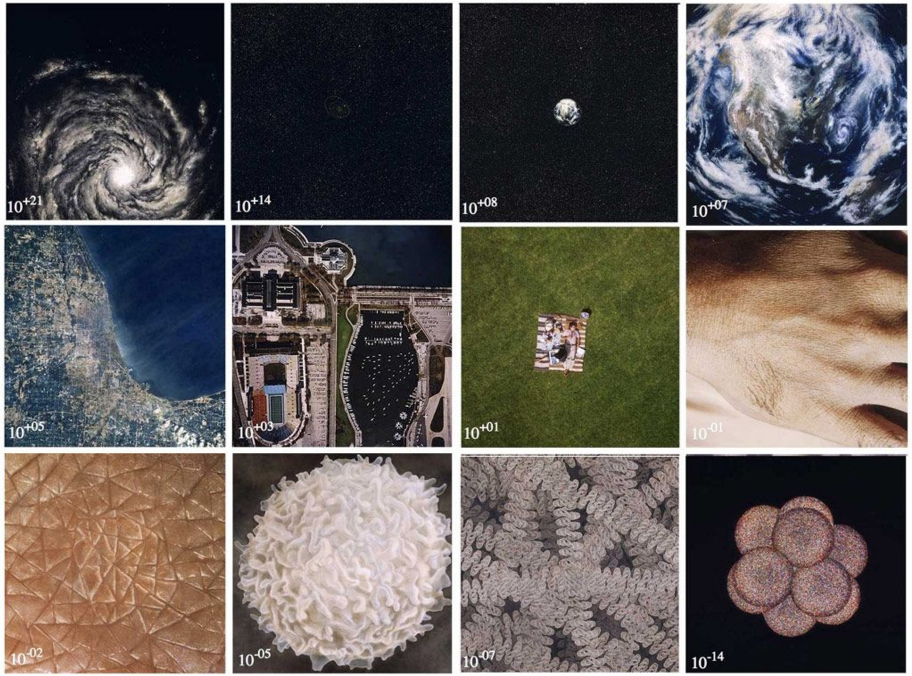
Lecture 6: Skill learning

PSYCH/COGS 140M 2026.04.23

Skill learning

Notes

- Lectures 6-8 cover *non-declarative learning*
    - 6: *Skill learning* - **High-level**: What practices and conditions help us build skills?
    - 7: *Habitization, sensitization, familiarization, and priming* **- More detailed**: What are the component mental and physical processes that comprise skills?
    - 8: *Conditioning* **- Most detailed**: Formally capture the building blocks of procedural learning, building towards a mathematical model.



Outline

- Defining skill learning
- Learning through practice
- Brain substrates

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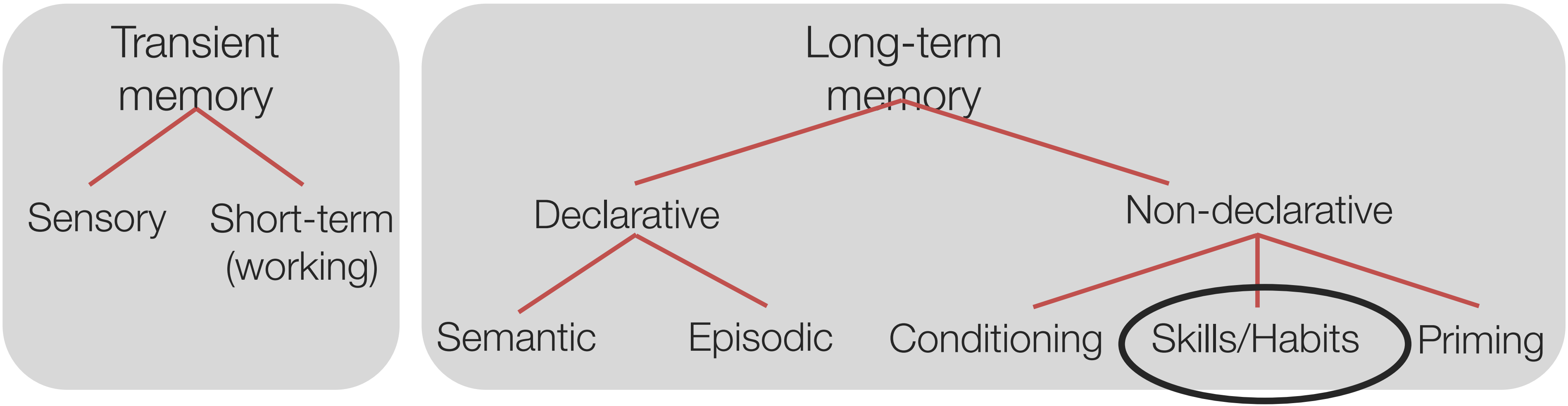
Multiple memory systems



- Acquisition of new skills and habits with practice over time
- Note: if you hear the term **procedural learning**, it **means the same thing as skill learning**!

Skill learning

Improved performance on motor, perceptual, or cognitive tasks with practice

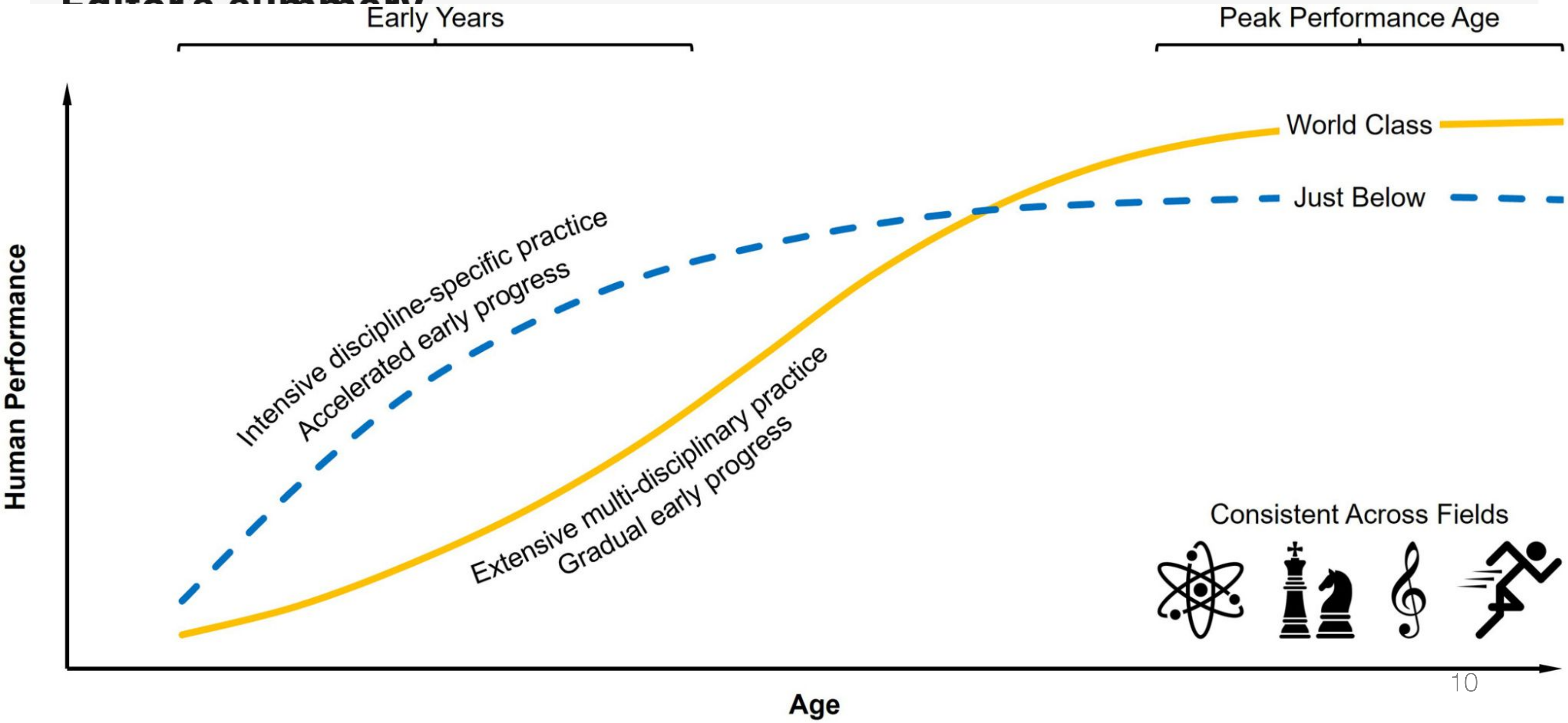
- Perceptual-motor skills: motor patterns guided by sensory inputs

52 x 17

- Cognitive skills: ability to problem solve and apply strategies

- Many skills involve a combination of motor, perceptual, and cognitive abilities

Editor's summary



Types of perceptual-motor skills

- Closed skills
  - Performing pre-defined sequences of actions
  - Gymnastics, ice-skating, or diving routines; playing a violin concerto
- Open skills
  - Require adjustments based on changes in the environment
  - Playing basketball; playing improvisational jazz
- Most skills fall somewhere on a spectrum of closed to open

Closed skill

Open skill


OLYMPUS
CHASE
US OPEN
CHASE
OLYMPUS
CITIZEN
IBM
96
MPH
usopen.org
Heineken
1 FEDERER
4 DJOKOVIC
7 7 6
6 5 5
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Stages of skill learning

- Fitts' three-stage model (1964)

- Cognitive stage
  - Initial period, typically verbal, in which effort is required to perform skill

- Associative stage
  - Less reliance on verbal rules
  - More stereotyped behavior

- Autonomous stage
  - Skill is automatic and requires little attention (habitual)

Stages of skill learning

- Learning to drive a car with manual transmission

- Cognitive stage
  - Clumsy and slow; rules rehearsed verbally
  - Need conscious effort and control
- Associative stage
  - Many actions (i.e., shifting) have become stereotyped
  - Often need conscious control to determine appropriate sequence of actions
- Autonomous stage
  - Driving requires little attention; can listen to the radio and chat with friends
  - May even realize that you don't remember your drive home

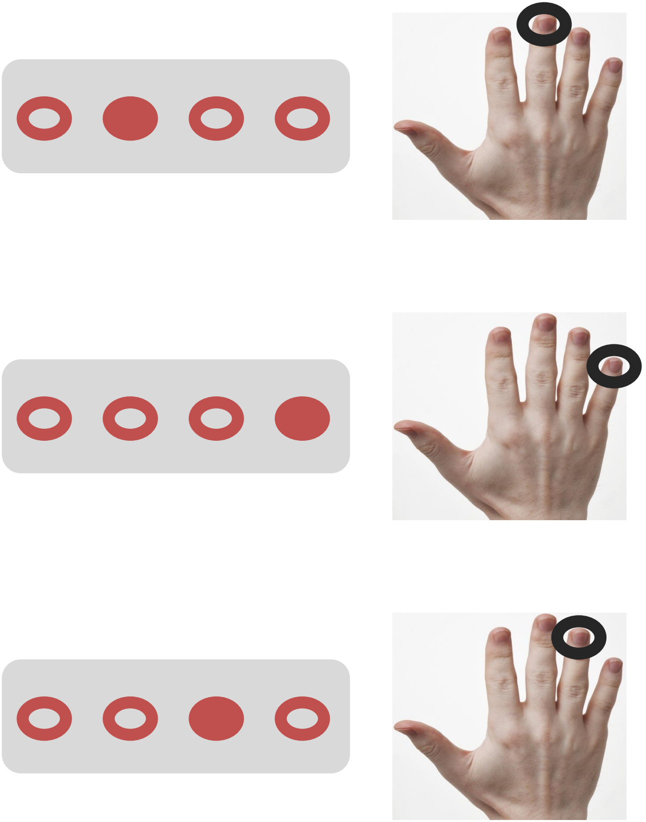
How is skill learning unique?

- Differences from declarative memory:
  - Can be acquired implicitly, i.e., without conscious awareness
  - Typically hard to verbalize
  - Requires practice

Implicit learning

Serial reaction time (SRT) task

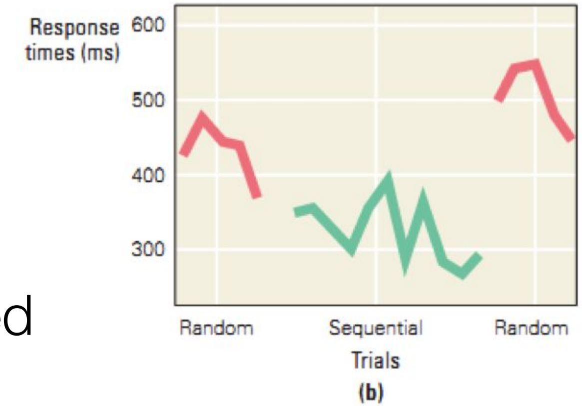
- Press button that corresponds to visual cue
- Cues mostly appear in random order, but some sequences repeat
- With training, reaction time to repeated sequence is faster than random sequences



Implicit learning

Serial reaction time (SRT) task

- Press button that corresponds to visual cue
- Cues mostly appear in random order, but some sequences repeat
- With training, reaction time to repeated sequence is faster than random sequences
- Typically, participants are unaware of repeating sequences!

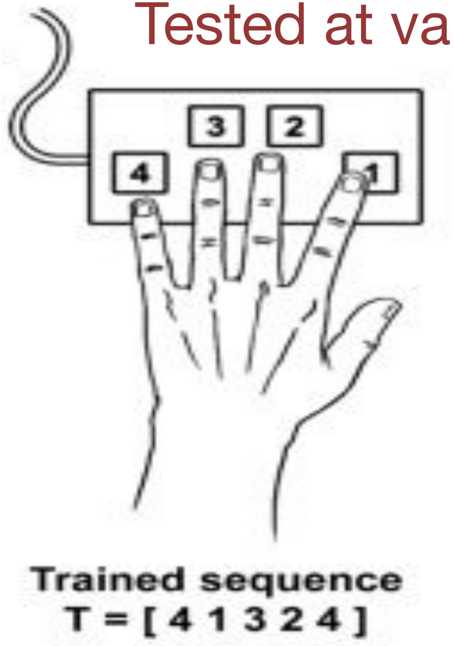


Skill Learning and Sleep

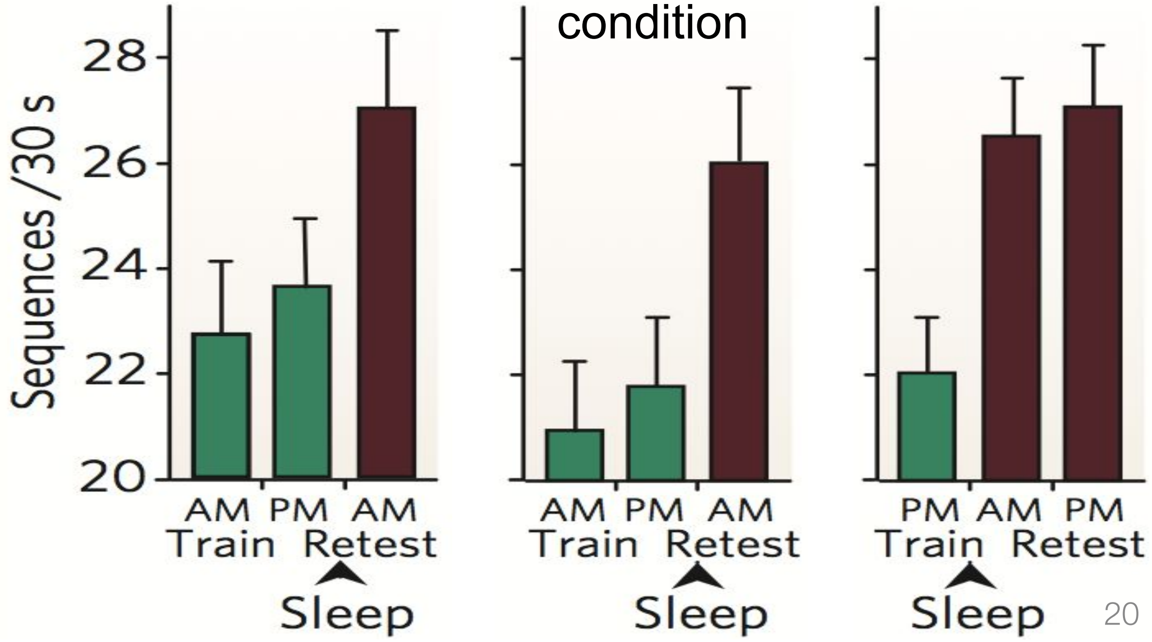
- One of the best ways to get better at a skill is to ……
- Sleep!

Skill Learning and Sleep

Subjects trained to learn sequence of key presses.
Tested at various delays either with sleep or wake in between.



“Hand rest” condition



Walker et al., 2002

Outline

- Defining skill learning
- Learning through practice
- Brain substrates

Practice

- Many factors influence how effective practice is

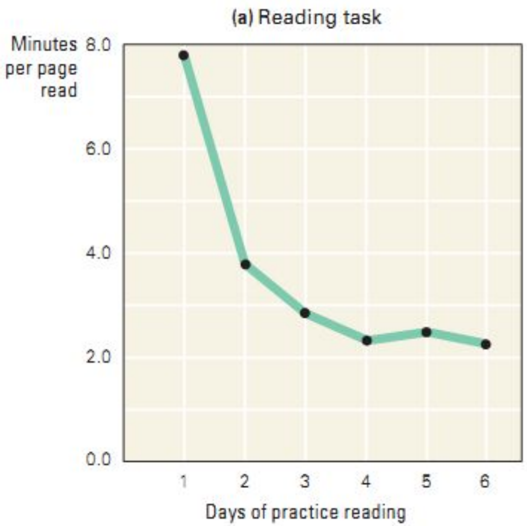
- Feedback
  - Knowledge of performance during practice

- Spacing
  - Massed: concentrated practice
  - Spaced: practice spread out over multiple sessions

- Variability
  - Constant: practice focused on a single skill
  - Variable: practice that alternates between a set of skills

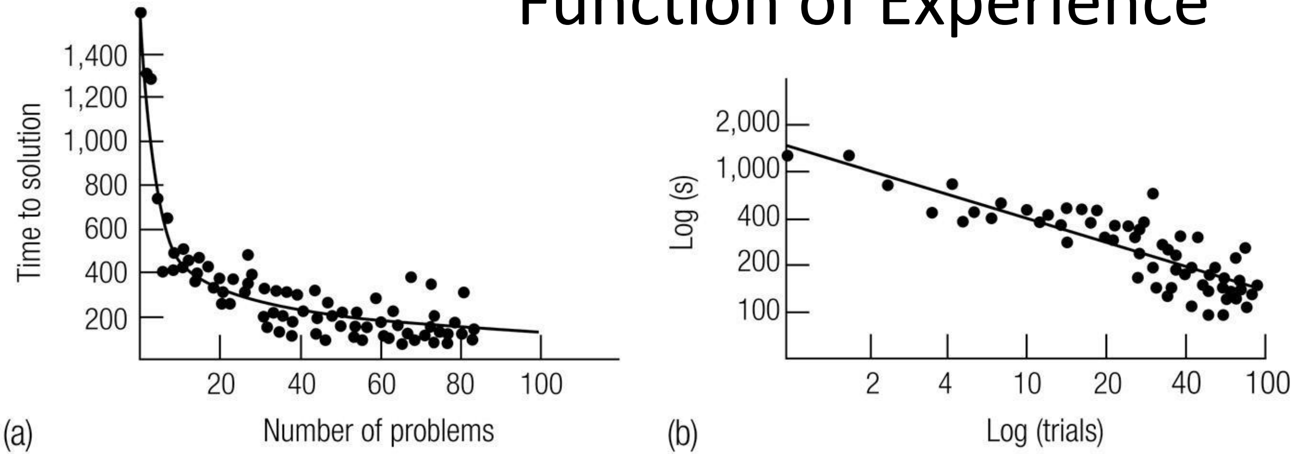
Power law of learning

- Gains in learning are very rapid at first, but rate of learning declines with practice
- Pattern holds true across a wide range of tasks (perceptual-motor, cognitive) and species



Geometry Proofs

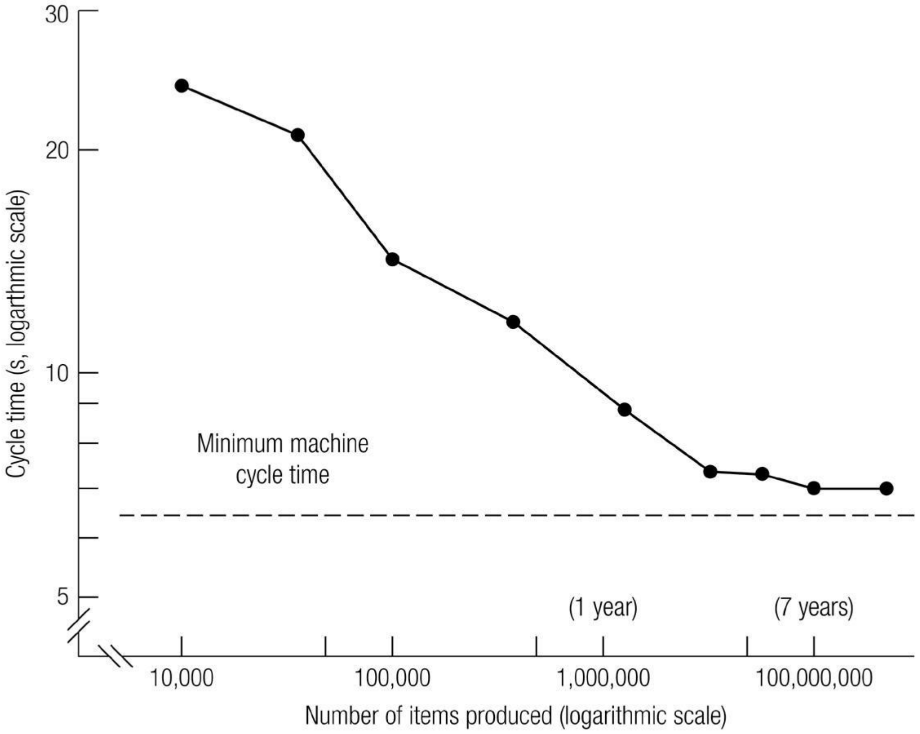
Time to Generate Geometry Proofs as a Function of Experience



Neves & Anderson (1981)

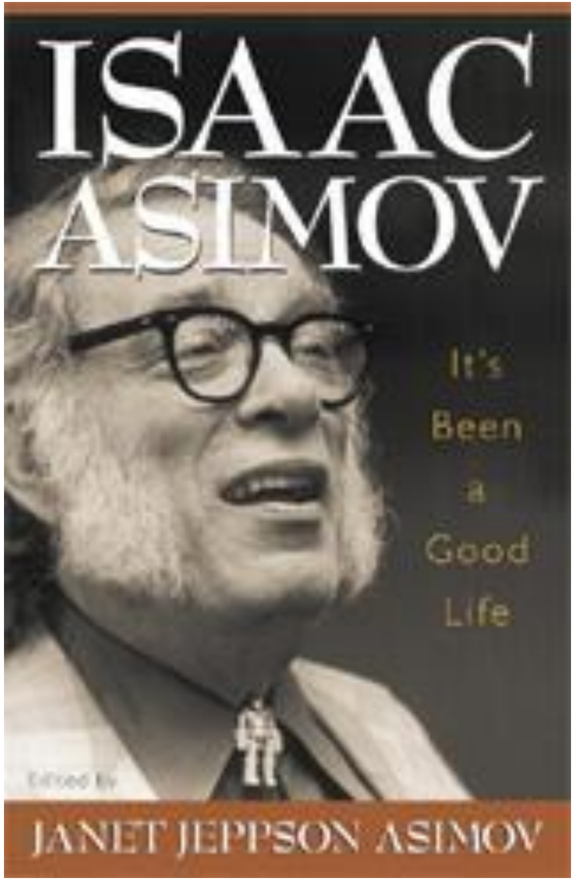
Cigar Production

Time to Produce a Cigar as a Function of Experience



Crossman (1959)

Writing Books

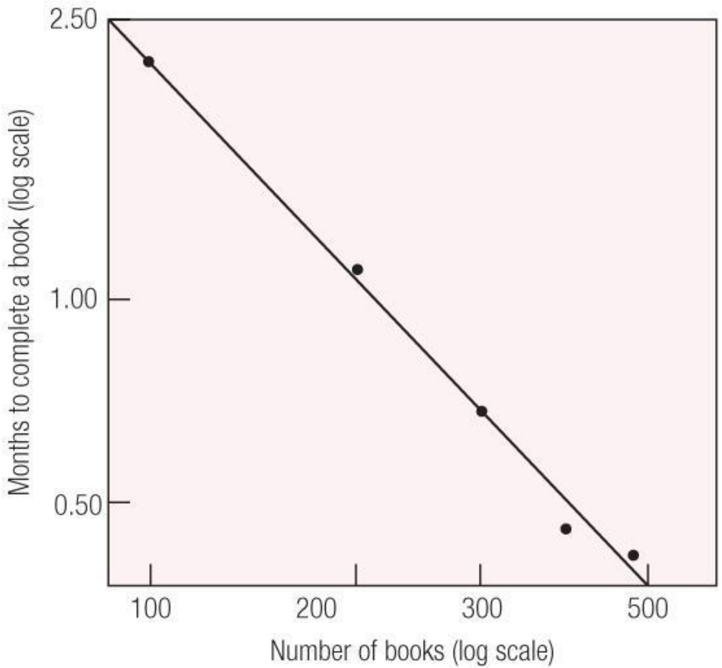


Isaac Asimov

- He loved to write and wrote every day 7:30am - 10 pm.
- Ohlsson (1992) treated Asimov's writing career as an experiment in learning how to write books.
- Plotted number of books (close to 500 books total) as a function of time

Isaac Asimov

# Books Total	*Cumulative Months*
100	237 months
200	350
300	419
400	465
490	507



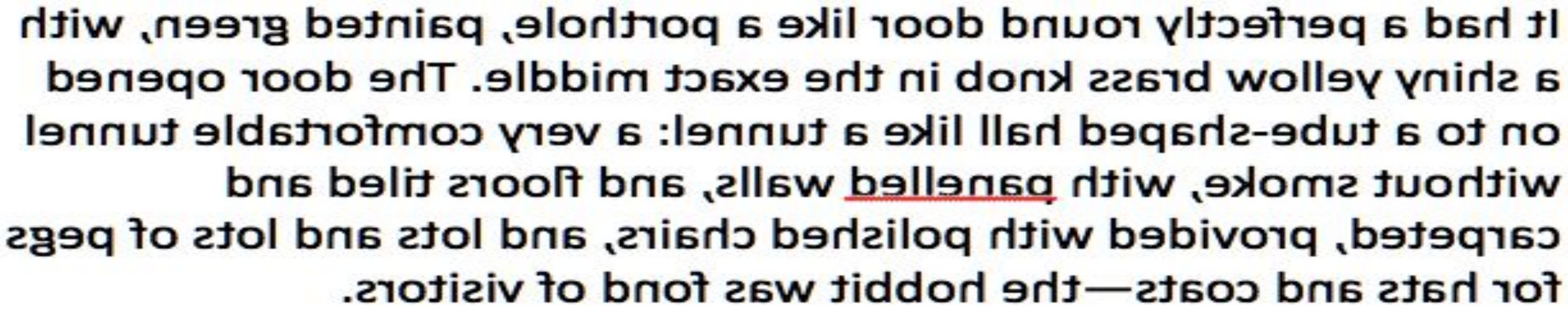
Ohlsson (1992)

Example: mirror-reversed reading

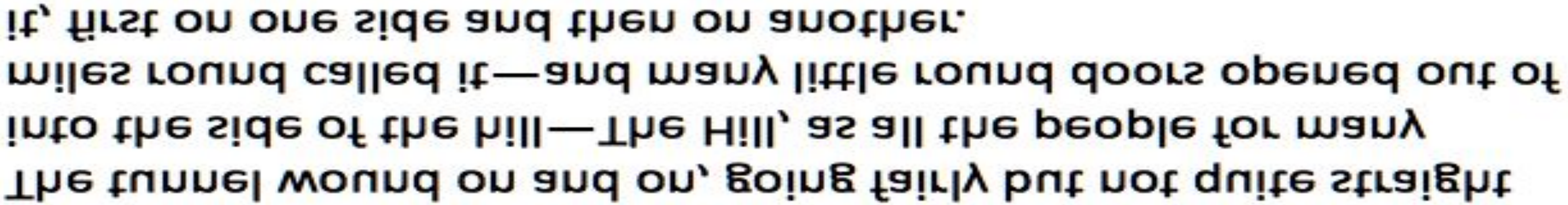
In a hole in the ground there lived a hobbit. Not a nasty, dirty, wet hole, filled with the ends of worms and an oozy smell, nor yet a dry, bare, sandy hole with nothing in it to sit down on or to eat: it was a hobbit-hole, and that means comfort.

Example: mirror-reversed reading

Mirror reversed



Mirror reversed and upside down



Example: mirror-reversed reading

- With mirror reading training, brain activity:
  - increases in basal ganglia
  - decreases in hippocampus

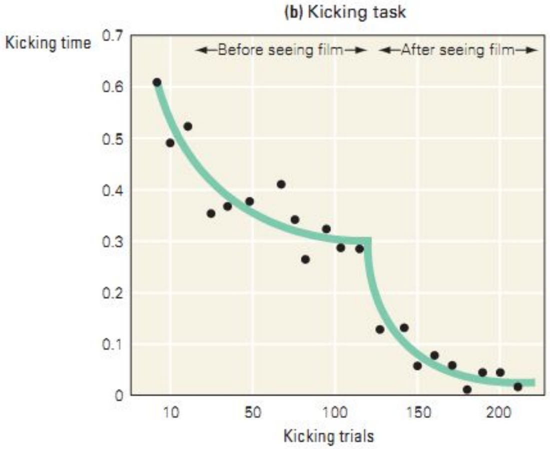
- Suggests switch from cognitive stage, which is reliant on declarative knowledge to associative stage, which is more reliant on procedural knowledge

Poldrack & Gabrieli (2001)

Overcoming the power law

Additional sources of feedback can reset the power law

E.g., Subjects perform “kicking task” to master kicking technique. First learn through visual feedback, then watch an instructive video



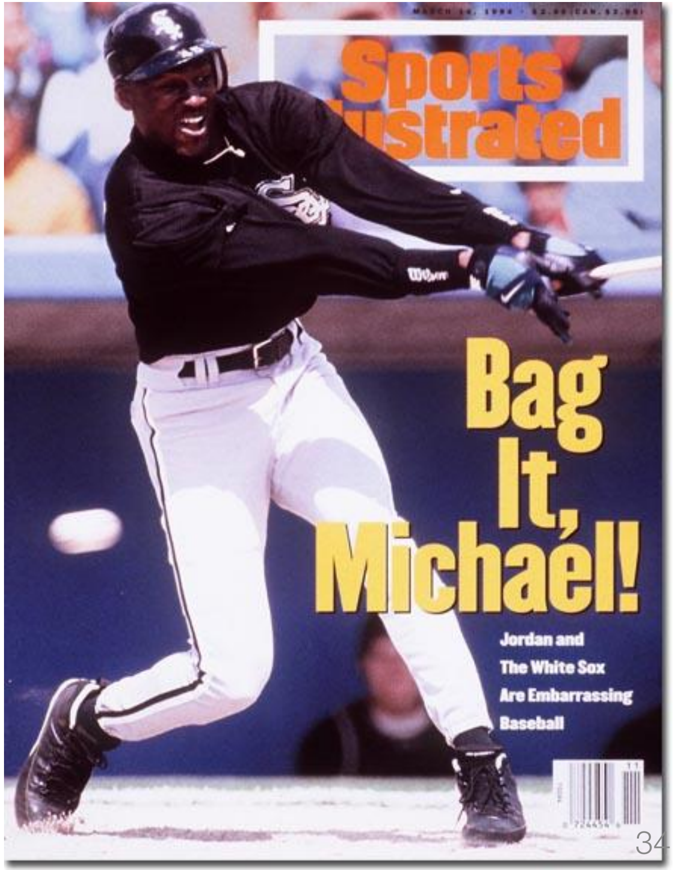
Hatze (1976)

Transferring skills

- Transfer: generalization of skill learning from one context to another
  - Example: Gymnast joining the diving team
- Do skills transfer?
- Generally, most skills do *not* transfer well

Transferring skills

'This Must Be Some Mistake,' Says Snowboarder After Winning Olympic Gold In Skiing



Transferring skills

- When are skills likely to transfer?
- Thorndike's identical elements theory (1901)
  - Transfer depends on the similarity of the training context to the new context
  - The more shared (identical) elements, the better the transfer of skill learning
- Theory supports the notion of transfer specificity
  - I.e., transfer is specific to instances in which two skills share the same elements

Becoming an Expert

Genuine expertise should pass three tests:

– Consistently superior performance relative to one's peers

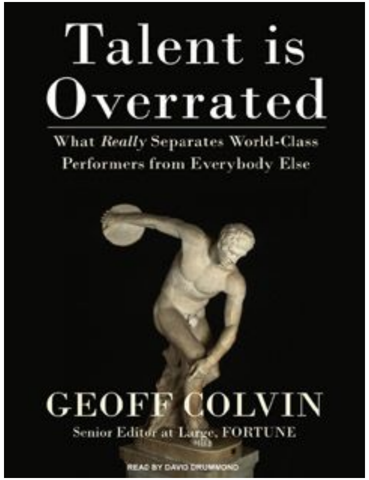
– Concrete results, e.g., winning more games

– Can be measured *and* replicated in the lab

10,000 hours

Research by Ericsson and others suggests that ~10,000 hours of deliberate practice are required to become an expert

- Musicians
- Athletes
- Medical doctors
- Chess masters

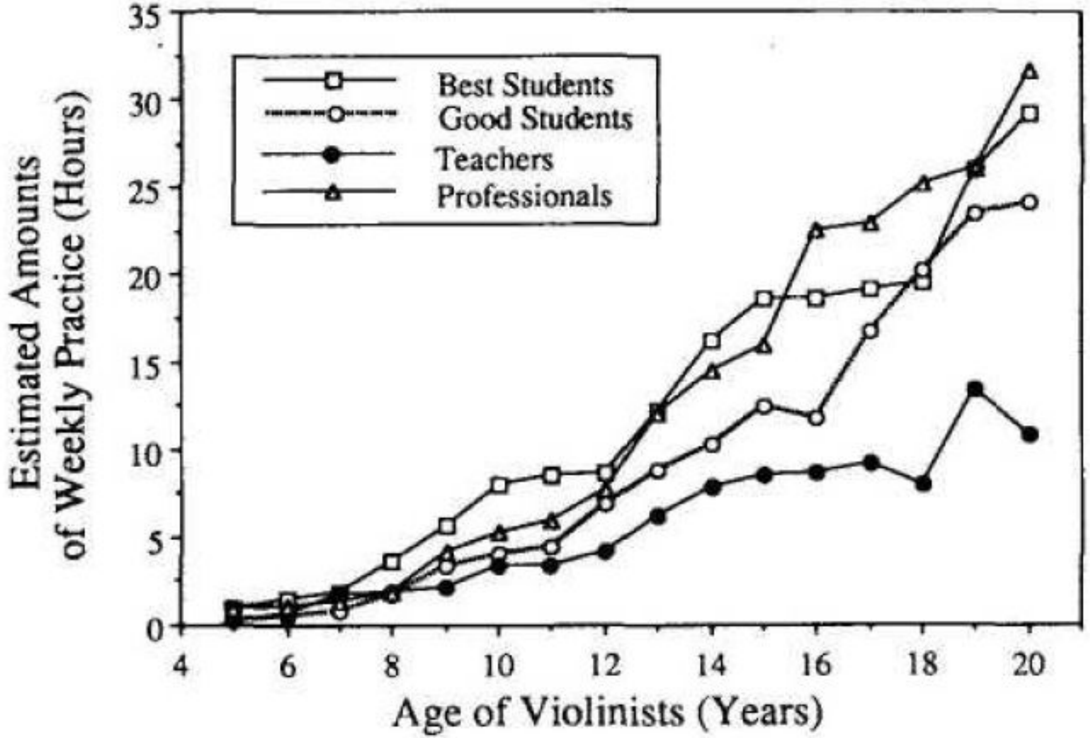


- Spacing is important – best to practice 2-3 hours per day for many years

Music: Talent or Practice?

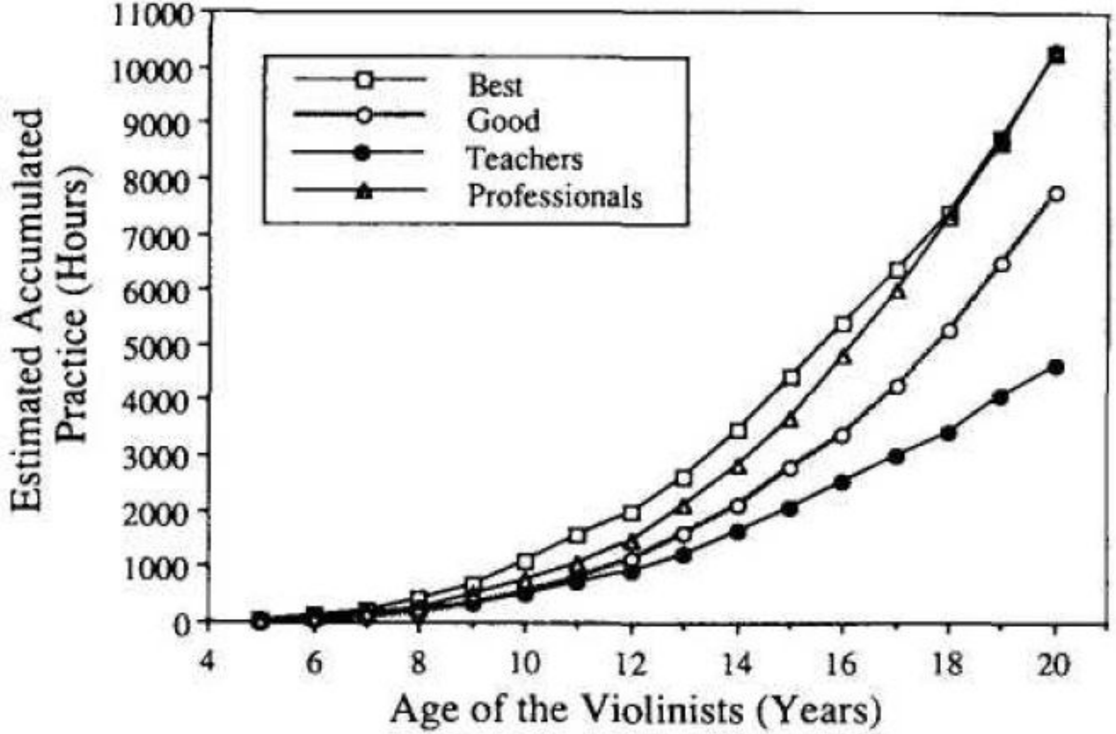
- Ericsson et al. (1993) studied background of musicians in terms of achievement level
- For weeks, each musician recorded hourly activity, including practice, leisure, and sleep.
- Based on these data, the researchers estimated accumulated practice over several years

Weekly practice: Violinists



Ericsson et al. (1993)

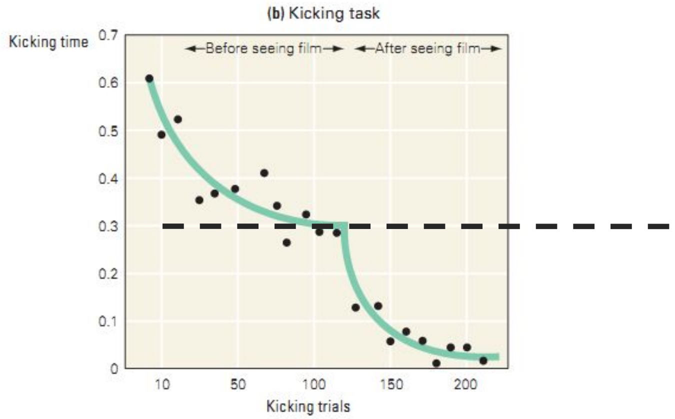
Estimated accumulated practice



Ericsson et al. (1993)

Deliberate practice

- Requires focused attention
- Requires feedback
- Requires regularly changing context and conditions



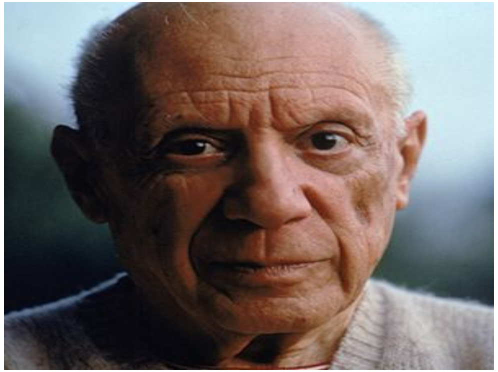
"Pretty good" plateau

Importance of failure

- Practice should be challenging
  - To get past a plateau, there must be a risk of failure
  - *Think of failure as another **desirable difficulty**!*
- Lacking awareness or not addressing failure are often reasons that we don't improve with practice
  - Practice doesn't necessarily make perfect!

Importance of failure

“I am always doing that which I cannot do, in order that I may learn how to do it.”



Pablo Picasso

“I've missed more than 9000 shots in my career. I've lost almost 300 games. 26 times, I've been trusted to take the game winning shot and missed. I've failed over and over and over again in my life. And that is why I succeed”

Michael Jordan

Necessity of 10,000 hours

- Do we truly need 10,000 hours of deliberate practice to become an expert?
- Age at which practice begins has a large influence on the time required to master a skill
- Some skills have less competition than others
  - Tip: If you want to be an expert without putting in the hours, choose an obscure skill!

Sufficiency of 10,000 hours

- Not everyone reaches the same level of expertise after the same amount of practice

- Why do some individuals acquire skills more rapidly than others?

  - “Innate” traits (e.g., height)
    - *But there are many “short” basketball players*

  - Motivation, desire, self-discipline

Pushback against 10,000 hours

- Many researchers criticize the idea that the amount of deliberate practice can fully explain differences in performance

- MacNamara et al, 2014

  – For a given domain (e.g., music), evaluated the relationship between performance and amount of deliberate practice

  – Question: how much of the variability in performance can be explained by variability in practice?

MacNamara et al, 2014

Amount of deliberate practice explained a relatively small amount of variability in performance

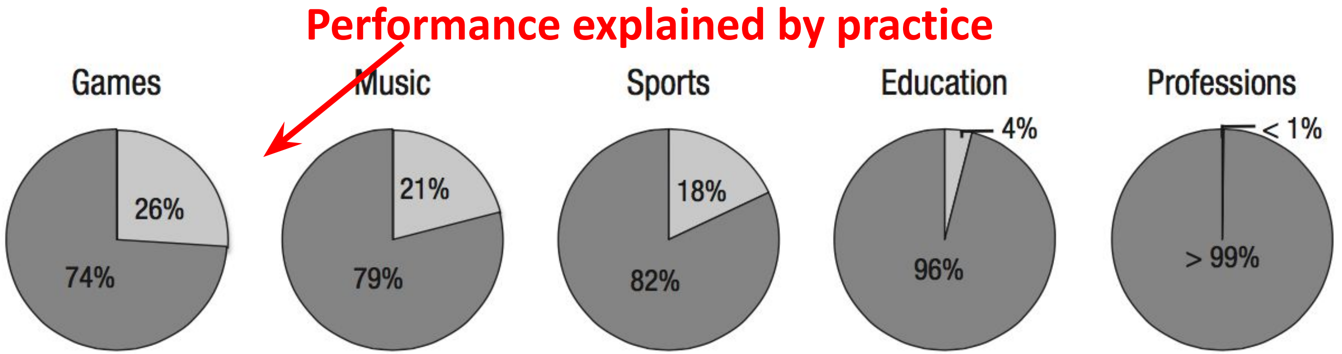


Fig. 3. Percentage of variance in performance explained (light gray) and not explained (dark gray) by deliberate practice within each domain studied. Percentage of variance explained is equal to $r^2 \times 100$.

The relationship between deliberate practice and performance differed across domains

MacNamara et al, 2014

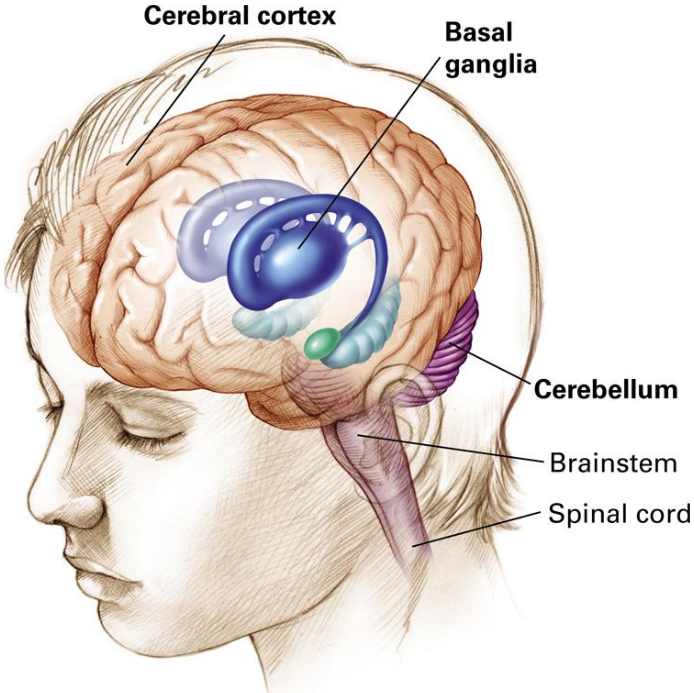
- Practice is a better predictor of performance in domains that are highly predictable
  - Predictable: running
  - Unpredictable: handling aviation emergency

Outline

- Defining skill learning
- Learning through practice
- Brain substrates

Brain substrates of skill learning

- Basal ganglia
- Cortex
- Cerebellum



Brain substrates of skill learning

- **Basal ganglia** plays a key role in regulating the direction, speed, and strength of movements
  - However, also important for cognitive skill learning!

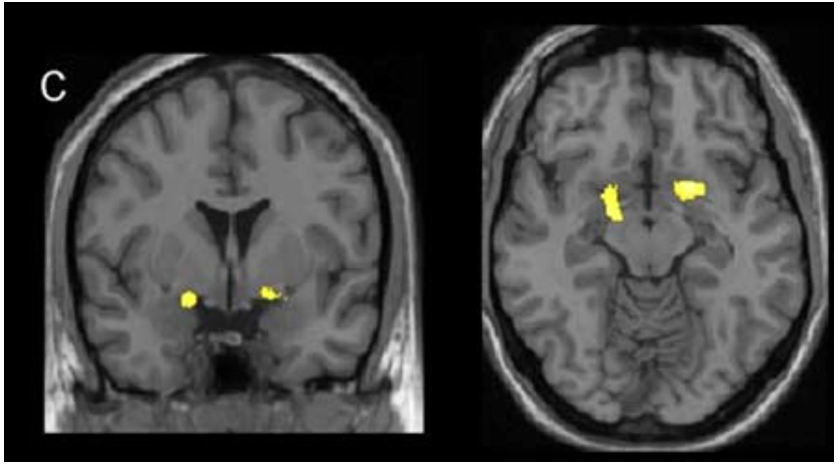
- Skills involve multiple components (sensory, motor, cognitive)
  - **Cortical areas** associated with these components change with practice

- **Cerebellum** is involved in regulating precise timing and aim of movements

Basal Ganglia

Motor skill learning: a good example is juggling—requires learning a specific sequence of motor movements.

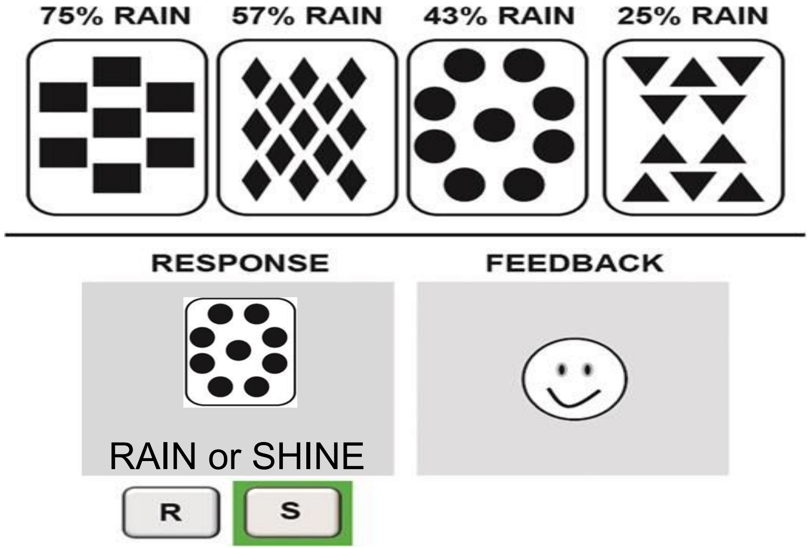
Several weeks of juggling increases grey matter in the basal ganglia.
But goes away when juggling practice stops!

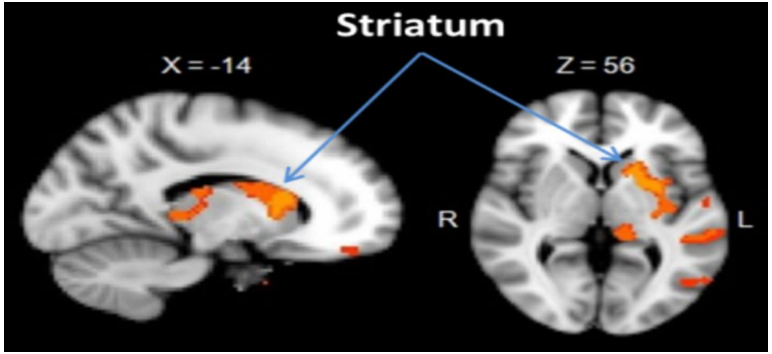


Boyke et al., 2008

Basal Ganglia

Weather prediction task: Predict ‘weather’ from cards that are probabilistically associated with rain vs. sun. As subjects improve at predicting outcomes (the weather), basal ganglia activation increases. Form of *cognitive skill learning*

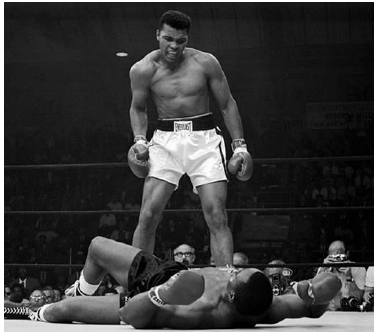




The striatum is part of the basal ganglia

Parkinson’s disease (PD)

- Neurodegenerative disease affecting about 1 million people in the US, typically later in life
- Associated with gradual death of neurons in the substantia nigra, which provides input to the basal ganglia
  - Decreased input causes dysfunction of the Basal Ganglia



- Cause is largely idiopathic, i.e., unknown
  - Genetic factors, particularly for early-onset PD
  - Environmental factors, e.g., head injuries or toxins

Parkinson's disease (PD)

- Primarily a movement disorder:
  - Muscle tremors
  - Rigidity and slowness of movement
  - Difficulty initiating movement

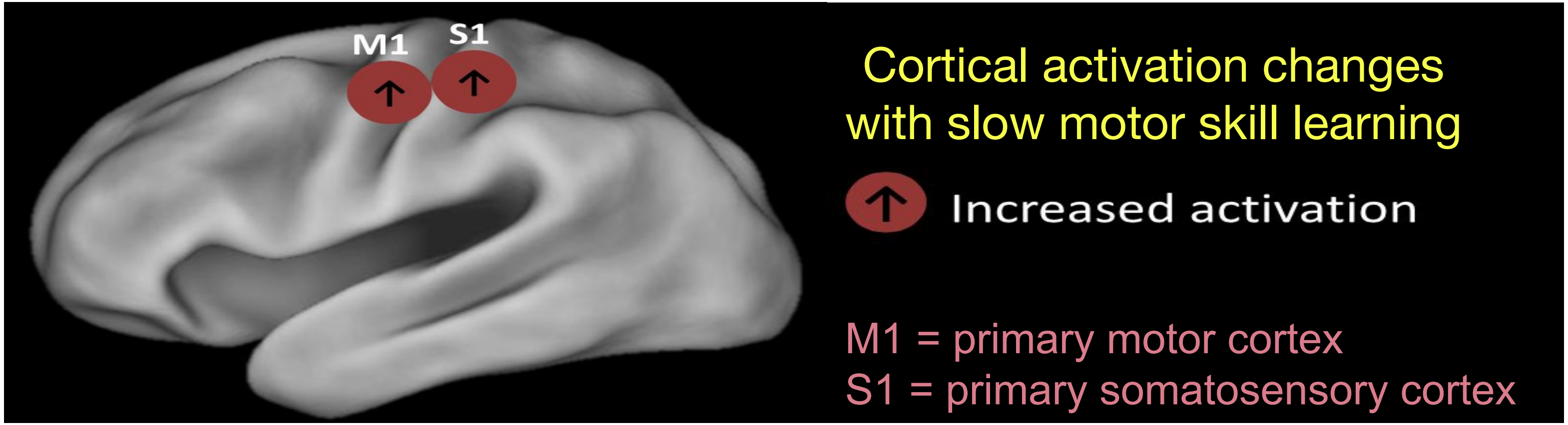
- But also affects cognition:
  - Impairments in skill learning (both perceptual-motor and cognitive)
  - Includes tasks like the weather prediction task

Sensory-motor areas

Skill learning results in cortical plasticity. Plasticity can either manifest in functional or structural changes in cortical areas.

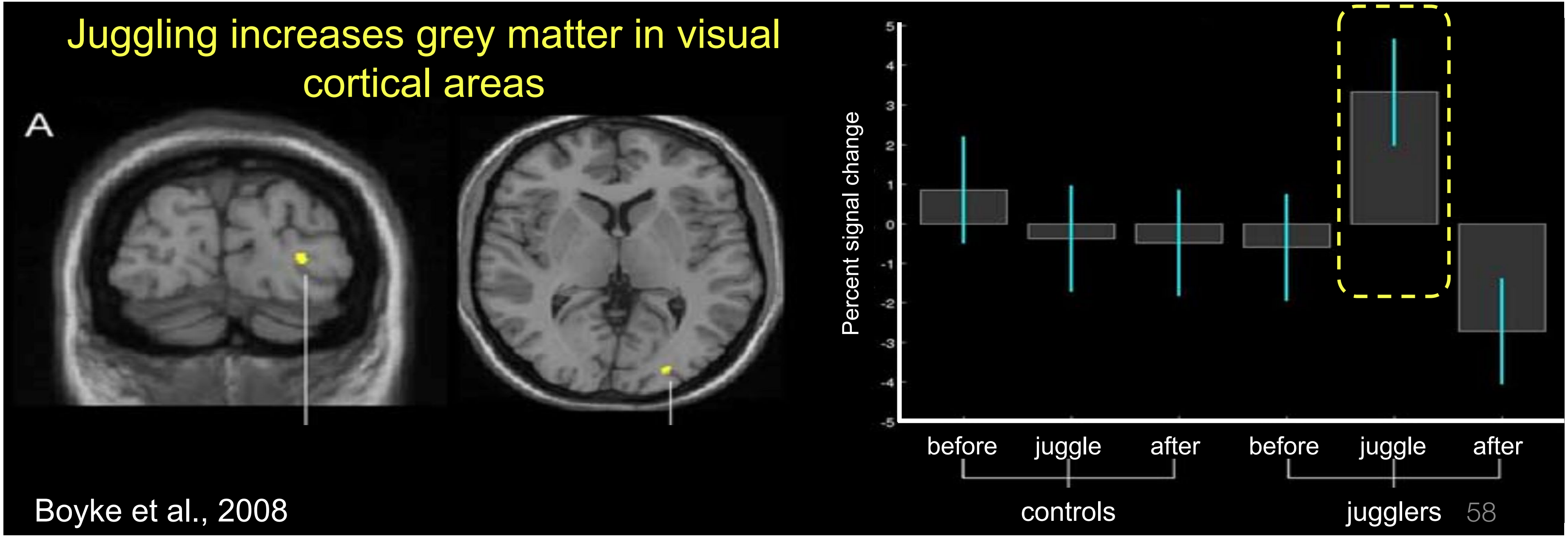
Functional changes

Repeated practice of a motor skill leads to increased *activation* in sensory areas



Structural changes

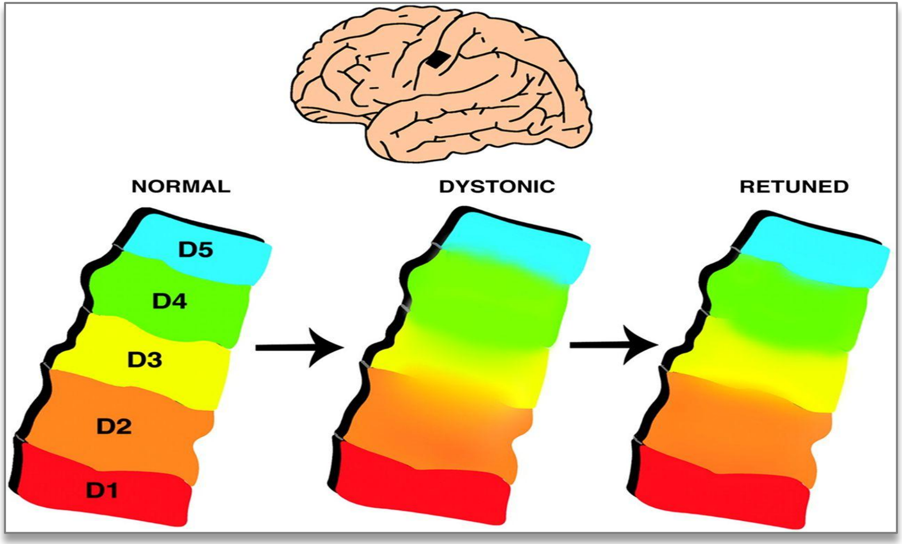
Skill learning can produce cortical expansion, or an increase in the amount of cortex dedicated to the learned skill



Musician's Dystonia

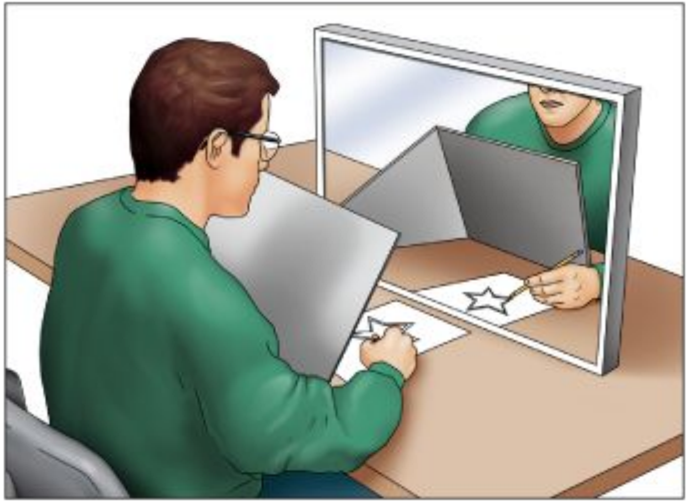
In some individuals, extensive skill learning can result in focal dystonia: a neurological disorder involving involuntary muscle movements

Musician's dystonia is one example. It is thought to occur when cortical expansion of a specific motor area 'spreads' into neighboring motor areas.



Cerebellum

Mirror tracing: Patients with damage to the *cerebellum* exhibit impaired mirror tracing learning (a form or perceptual-motor skill learning)



Summary: Brain substrates of skill learning

- Basal ganglia involved in motor *and* cognitive skill learning
- Motor skill learning also involves sensory-motor areas and cerebellum
  - Cortical plasticity occurs with learning: may reflect activation changes (functional) or structural changes
- Relevant sensory areas (e.g., visual cortex) may also exhibit training-induced plasticity

Next time…

- Habitization